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halox: Dark matter halo properties and large-scale structure calculations using JAX

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Software

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Summary

Dark matter halos are fundamental structures in cosmology, forming the gravitational potential wells hosting galaxies and clusters of galaxies. Their properties and statistical distribution (including the halo mass function) are invaluable tools to infer the fundamental properties of the Universe. The `halox` package is a JAX-powered Python library enabling differentiable and accelerated computations of key properties of dark matter halos, and of the halo mass function. The automatic differentiation capabilities of `halox` enable its usage in gradient-based workflows, *e.g.* in efficient Hamiltonian Monte Carlo sampling or machine learning applications.

Statement of need

In cosmology and astrophysics, modeling dark matter halos is central to understanding the large-scale structure of the Universe and its formation. This has motivated the development of many toolkits focused on halo modeling, such as, *e.g.*, halofit (Smith et al., 2003), halotools (Hearin et al., 2017), colossus (Diemer, 2018), or pyCCL (Chisari et al., 2019). Recently, the AI-driven advent of novel computational frameworks such as JAX (Bradbury et al., 2018), have led to the development of differentiable and hardware-accelerated software to simulate and model physical processes, with *e.g.* Brax (Freeman et al., 2021) and JAX, MD (Schoenholz & Cubuk, 2020). The increasing complexity of cosmological data and astrophysical models has motivated the wide adoption of this framework in cosmology, where JAX-powered software has been published to address a wide variety of scientific goals, including modeling fundamental cosmological quantities, with, *e.g.*, JAX-cosmo (Campagne et al., 2023) and LINX (Giovanetti et al., 2024); simulating density fields and observables, with, *e.g.*, SHAMNet (Hearin et al., 2022), DISCO-DJ (Hahn et al., 2024), JAXpm (Differentiable Universe Initiative, 2025), and JAX-GalSim (JAX-GalSim developpers, 2025; Mendoza et al., 2025); emulating likelihoods for accelerated inference, with, *e.g.*, CosmoPower-JAX (Piras & Spurio Mancini, 2023) and candl (Balkenhol et al., 2024); or modeling various physical properties of dark matter halos, such as mass acretion history (Diffmah, Hearin et al., 2021), galaxy star formation history (Diffstar, Alarcon et al., 2023), halo concentration (Diffprof, Stevanovich et al., 2023), gas-halo connection (picasso, Kéruzoré et al., 2024), and halo mass function (Buisman et al., 2025)[1].

The `halox` library offers a JAX implementation of some widely used properties which, while existing in other libraries focused on halo modeling, do not currently have a publicly available, differentiable and GPU-accelerated implementation, namely:

- Radial profiles of dark matter halos following a Navarro-Frenk-White (NFW, Navarro et al., 1997) distribution;
- The halo mass function, quantifying the abundance of dark matter halos in mass and redshift, including its dependence on cosmological parameters;

[1] Note that halox also provides an implementation of the halo mass function, but choses a lighter, halo model-based approach; see **Features**.

- The halo bias.

The use of JAX as a backend allows these functions to be compiled and GPU-accelerated, enabling high-performance computations; and automatically differentiable, enabling their efficient use in gradient-based workflows, such as sensitivity analyses, Hamiltonian Monte-Carlo sampling for Bayesian inference, or machine learning-based methods.

Features

Available physical quantities

The `halox` library seeks to provide JAX-based implementations of common models of dark matter halo properties and of large-scale structure. At the time of writing (software version 1.1.0), this includes the following properties:

- Cosmological quantities: `halox` relies on JAX-cosmo (Campagne et al., 2023) for cosmology-dependent calculations, and includes wrapper functions to compute some additional properties, such as critical density ρ_c and differential comoving volume element $\mathrm{d}V_c/\mathrm{d}\Omega\mathrm{d}z$.
- Radially-dependent physical properties of NFW dark matter halos. Our implementations are based on the analytical derivations of Łokas & Mamon (2001), and include the following quantities:
  - Matter density $\rho(r)$;
  - Enclosed mass $M(\leq r)$;
  - Gravitational potential $\phi(r)$;
  - Circular velocity $v_\mathrm{circ}(r)$;
  - Velocity dispersion $\sigma_v(r)$;
  - Projected surface density $\Sigma(r)$.
- Large-scale structure: Building upon the power spectra computations implemented in JAX-cosmo, `halox` provides implementations of the RMS variance of the matter distribution in spheres of radius R, $\sigma(R)$. It also includes a wrapper function to perform the computation within the Lagrangian radius of a halo of mass M, $\sigma(M)$.
- Halo mass function (HMF): The HMF model of Tinker et al. (2008), predicting $\mathrm{d}N/\mathrm{d}\ln M$ as a function of halo mass M, redshift z, and cosmology.
- Halo bias: The linear bias model of Tinker et al. (2010) as a function of halo mass M, redshift z, and cosmology.
- Overdensities: All properties in `halox` can be computed for spherical overdensity (SO) halo masses defined for any critical overdensity value. Convenience functions are provided to convert halo properties from one critical overdensity to another, or to convert critical overdensities to mean matter overdensities.

Automatic differentiation and hardware acceleration

All calculations available in `halox` are written using JAX and JAX-cosmo. As a result, all functions can be compiled just-in-time using `jax.jit`, hardware-accelerated, and are automatically differentiable with respect to their input parameters, including halo mass, redshift, and cosmological parameters.

Validation

All functions available in `halox` are validated against existing, non-JAX-based software. Cosmology calculations are validated against Astropy (Astropy Collaboration et al., 2022) for varying cosmological parameters and redshifts. Other quantities are validated against `colossus` (Diemer, 2018) for varying halo masses, redshifts, critical overdensities, and cosmological

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parameters. These tests are included in an automatic CI/CD pipeline on the GitHub repository, and presented graphically in the online documentation.

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